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Todor Hristov

Education

Sep 2021 – March 2025 — **Computer Science (MSc)**, *ETH Zürich*, Switzerland
- Overall Grade: 5.38/6.00
- Received the Jury Award in the Game Programming Laboratory course.
- Units studied include:
  - Advanced Systems Lab
  - Information Security Lab
  - Program Verification
  - Automated Software Testing
  - System Security
  - Principles of Distributed Computing
  - Design of Parallel and High-Performance Computing

Sep 2017 – July 2020 — **Computer Science (BSc)**, *University of Bristol*, UK
- **Recipient of The Hargrave Scholarship for Academic Achievement**
- Overall Grade: 81% (First Class Honours)
- Attended the Algorithms Research Reading Group.
- Units studied include:
  - Machine Learning
  - Computational Neuroscience
  - Data Structures and Algorithms
  - Concurrent Computing
  - Object-Oriented Programming
  - Software Product Engineering
  - An Introduction to High Performance Computing

Sep 2012 – June 2017 — **Secondary Education**, *High School of Mathematics "Dr Petar Beron"*, Bulgaria
- Overall Grade: 5.98/6.00 | Major: Informatics and Mathematics

Work Experience

June 2025 – Present — **Software Engineer**, *Huawei*, Zürich, Switzerland
- Added Model Context Protocol (MCP) support to the LLM-enabled CodeArts IDE.
- Designed and optimised algorithms that reduced the overall token usage by 43% to 52%.
- Added MCP UI components for tool call user information, approval buttons, and server monitoring.
- Validated the MCP functionality using MCP-Bench (`github.com/Accenture/mcp-bench`).
- Collaborated with cross-functional teams to integrate LLM features into existing IDE pipelines.
- Gained valuable experience in `TypeScript`, Prompt Engineering, and System Administration.

March 2023 – August 2023 — **Research Assistant**, *Oracle*, Zürich, Switzerland
- Contributed to the development of the Parallel Graph AnalytiX (PGX) platform.
- Designed and optimised algorithms for efficient computation on large, complex graphs.
- Implemented new features and enhancements for the PGX platform, leveraging `Java` and `Python`.
- Worked in a supportive, close-knit team to develop robust and scalable software solutions.

Sep 2020 – August 2021 — **Backend Developer**, *Inexogy Smart Metering (Discovergy GmbH)*, Heidelberg, Germany
- Developed a Smart Meter Gateway Administrator (SMGWA) for energy data management.
- Used `Go` for its development, optimising system performance and integration between components.
- Leveraged `Bash` for deployment and automation, streamlining the different processes.
- Gained hands-on experience in the full development life cycle.

June 2018 – August 2018 — **Systems Analyst Intern**, *VarSys Ltd.*, Varna, Bulgaria
- Diagnosed and repaired hardware and software issues, improving technical troubleshooting skills.
- Effectively communicated technical problems to non-technical clients.
- Researched Ethereum mining hardware and software for a client project as the primary lead.
- Assisted in client deployments, recommending solutions and documenting results.

Projects

2024 **Large Language Models for Email Phishing Training and Education**,
Prof. Dr. Srdjan Čapkun, Dr. Daniele Lain, ETH Zürich, Switzerland
- Master's Thesis with the System Security Group, focused on AI-enhanced email phishing training.
- Built a modular platform with `React`, `FastAPI`, and `MongoDB`, deployed in three `Docker` containers.
- Conducted a 201-participant study showing that LLMs enhanced training-page effectiveness.
- Available on ETH Zürich's Research Collection: `doi.org/10.3929/ethz-b-000695282`

2022 **Seth's Pyramall**, *Game Programming Laboratory*, ETH Zürich, Switzerland
- Developed a 2D platformer in `C#` using the `MonoGame` framework, as part of a six-person team.
- Recipient of the Jury Award, given by the expert game developers from Studio Gobo.
- Implemented a grid-based sand simulation, creating realistic flow and piling dynamics.
- YouTube Trailer: `youtube.com/watch?v=u3z3P-rHndg`

2020 **Improving Mouse Detection via Motion Tracking**,
Dr. Sion Hannuna, Dr. Neill Campbell, University of Bristol, UK
- Bachelor's Thesis focused on improving object detection through motion tracking.
- Applied SORT (tracking-by-detection) to reduce false positives and identify hard-to-detect frames.
- Performance evaluated against bootstrapping and classical data augmentation methods.

2020 `C++` **Raytracer**, *Computer Graphics*, University of Bristol, UK
- Developed a raytracer using Simple DirectMedia Layer (SDL) and OpenGL Mathematics (GLM).
- As part of a two-person team, implemented the following features:
  - Soft shadows
  - Anti-aliasing
  - Mirrors
  - Textures
  - Phong Shading
  - Glass

Achievements

2020 **Hargrave Scholarship for Academic Achievement**, *University of Bristol*, UK
- Awarded to the top final-year BSc Computer Science student for outstanding academic performance.

2018 – 2019 **Top 100 in the World**, *Bloomberg Global CodeCon Finals*, UK
- Represented the University of Bristol at Bloomberg's global coding finals in 2018 and 2019.
- Ranked Top 100 worldwide, solving eight advanced algorithmic problems in `C++`.

2018 **First Place from University of Bristol**, *AIGaming Mini-Hack*, UK
- Ranked first among University of Bristol participants in an AI game-agent competition.
- Developed a `Python`-based approximation to the NP-hard Travelling Salesman Problem.

2017 **First Place**, *Accessibility Hack 2017*, University of Bristol, UK
- Awarded first place in a group competition focused on creating solutions to enhance accessibility.
- Developed a Chrome Extension that detects and simplifies complex website text.
- GitHub Repository: `github.com/LukeStorry/simple_read`

2015 **Gold Medal**, *National Spring Competition in Informatics*, Bulgaria
- Gold medal (1st place) in one of Bulgaria's most prestigious national programming competitions.
- Solved three advanced `C++` problems under strict time and memory constraints.

Skills

Technical	Programming Languages:	`C/C++`, `Python`, `TypeScript`, `Java`, `Go`, `Bash`
	Parallel Computing:	`MPI`, `OpenMP`
	Software Development:	Test Driven Development, Agile Software Development
	Tools & Technologies:	`Git`, `Docker`, LaTeX
	Computer Science Fundamentals:	Data Structures and Algorithms, Computer Architecture
Languages	English (C1) | German (A2) | Bulgarian (Native)	

Hobbies

Playing table football and simulation video games, such as *Cities: Skylines* and *Minecraft*.